



Displax multitouch technology

Now showing...

3D displays. A new feature offered by several brands including ASUS, Acer, Sharp, Lenovo and MSI. Imagine seeing a 3D movie or playing a 3D game on an ultra-portable laptop! "MSI's all-new 3D notebook has a 39.6cm (15.6-inch) 120Hz 3D high-definition display. Its graphical processing unit and 120Hz 3D panel work together to create an average simultaneous presentation of 60 images for both the left and right eyes. With NVIDIA 3D Vision glasses, lifelike 3D images appear and create the perfect integration of real and virtual worlds," describes Sam Chern, marketing director-notebooks, MSI.

High-definition displays. While the high-definition multimedia interface (HDMI) has been the luxury of select high-end laptop models, Mukesh Srivastava, product manager-VAIO, Sony India, feels it is all set to become mainstream now. "As high-definition ability is being adopted in LCD televisions and cameras as well, this technology is all set to become pervasive across categories."

Low-voltage processors. Low-voltage processors such as Intel's new ultra-low-voltage Dual Core and Core2Duo processors are going to become part and parcel of laptops. These processors do not inhibit performance as such, and boost overall battery life to a great extent. Since these processors are also cooler, they enable thin, sleek designs.

Multi-touch touchscreens. Multi-touch capability allows the user to give complex commands using simple gestures or movements involving multiple fingers—say, a two-finger tap or a three-finger tap. Apple's new touch pads also support pinching and rotating. At the Consumer Electronics Show (CES) 2010, Lenovo displayed what is touted to be the industry's first multi-touch capacitive touchscreen netbook tablet, the IdeaPad S10-3t. This netbook can fold over and swivel 180 degrees to become a touch tablet. Soon after, in March, HP upgraded its EliteTouch 2740p tablet to a multi-touch capacitive display.

called Displax has developed a thin polymer sheet, embedded with a grid of nanowires, that can be stuck on any opaque or transparent surface to turn it into an interactive input device. It can even be used on liquid crystal display

(LCD), organic light-emitting diode (OLED) and E-Ink display.

A microcontroller processes the multiple input signals that it receives from the grid of nanowires. When the screen is touched, it causes an electri-

cal disturbance, which the microcontroller analyses to decode the location of each input on that grid. The film, which can currently detect up to 16 fingers on a 127cm (50-inch) screen, comes with the required firmware, driver, USB interface and a control panel for user calibration and settings. The first products based on this technology might be shipped sometime this July.

In a *Wired* story on this technology, Daniel Wigdor, a user experience architect for Microsoft who focuses on multi-touch and gesture-based computing, has suggested that if Displax can extend this technology to work on larger displays, it would lead to massive multi-touch. An interesting suggestion!

Power supply is another area of mobility where a lot of innovation is happening. Making the process of charging less cumbersome and enhancing the life of batteries are key areas of research as far as the industry is concerned.

Recently, MSI came out with a wireless notebook battery charger. "Just set a notebook on the wireless battery charging pad (a power cord is unnecessary) to automatically charge the notebook's battery and help make the high-tech lifestyle even easier and more convenient," says Sam Chern, marketing director-notebooks, MSI.

Powermat has a similar wireless charging mat that can charge three devices (mobile, tablet, MP3 player, camera, etc) in parallel. However, Powermat does not seem ready for laptop charging yet.

There is a lot happening out there, and most of it hints at the graduation of ultra-portability into uber-portability. Netpad (a netbook with retractable keypads), Kidget (an innovative educational gadget for kids), Entourage's eDGe eReader, HP's Slate... the list of innovations in this space is endless. ●

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